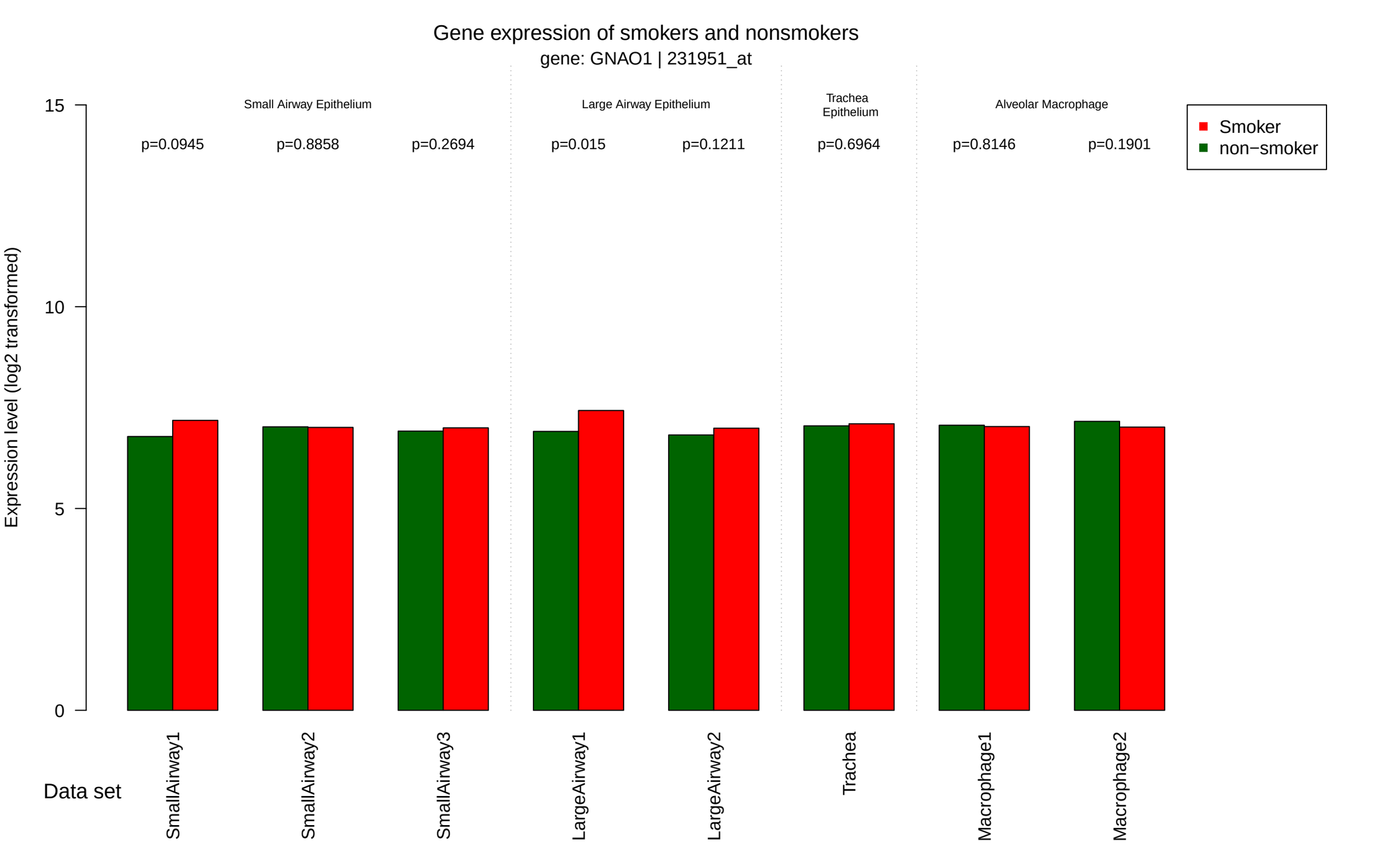
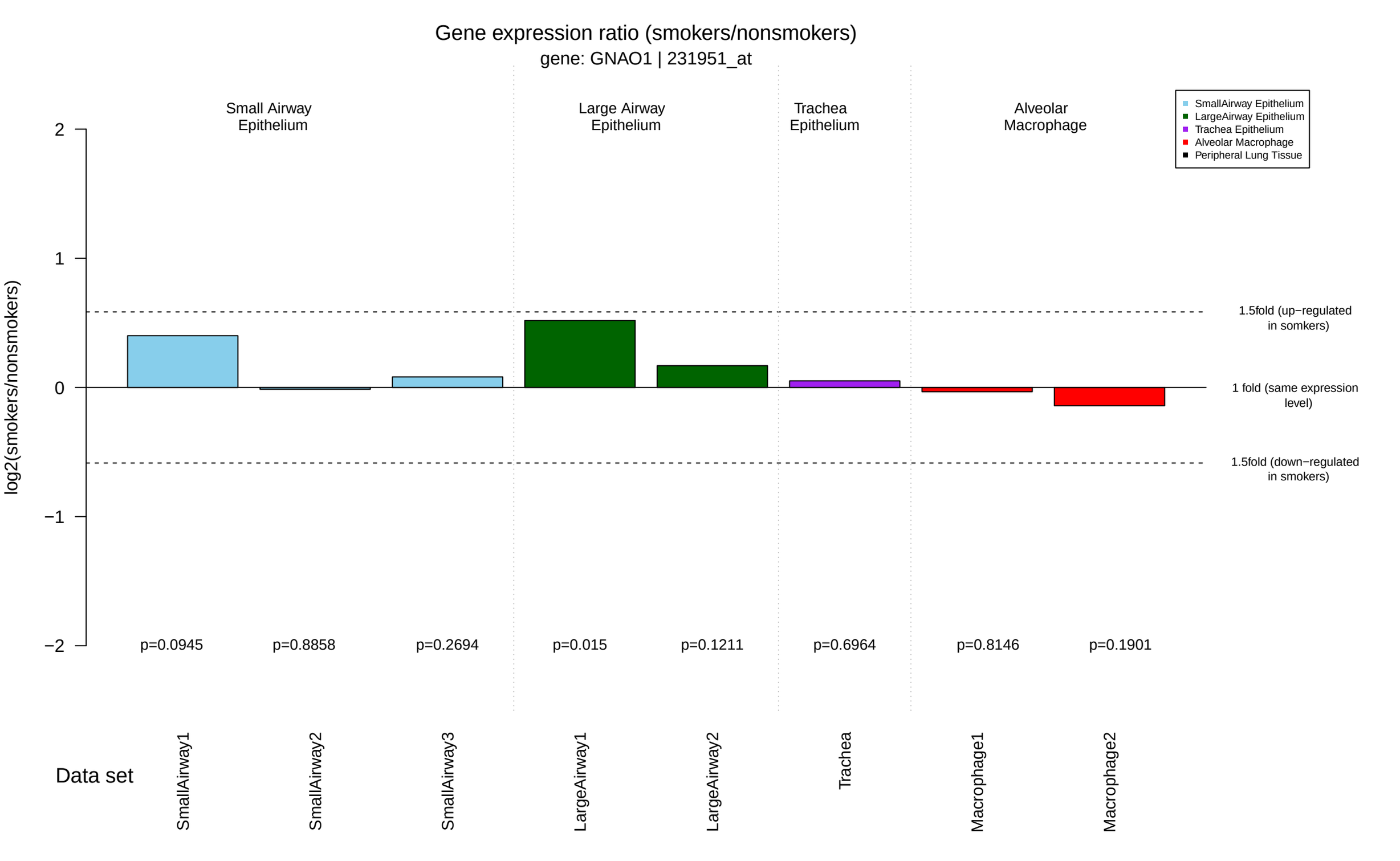
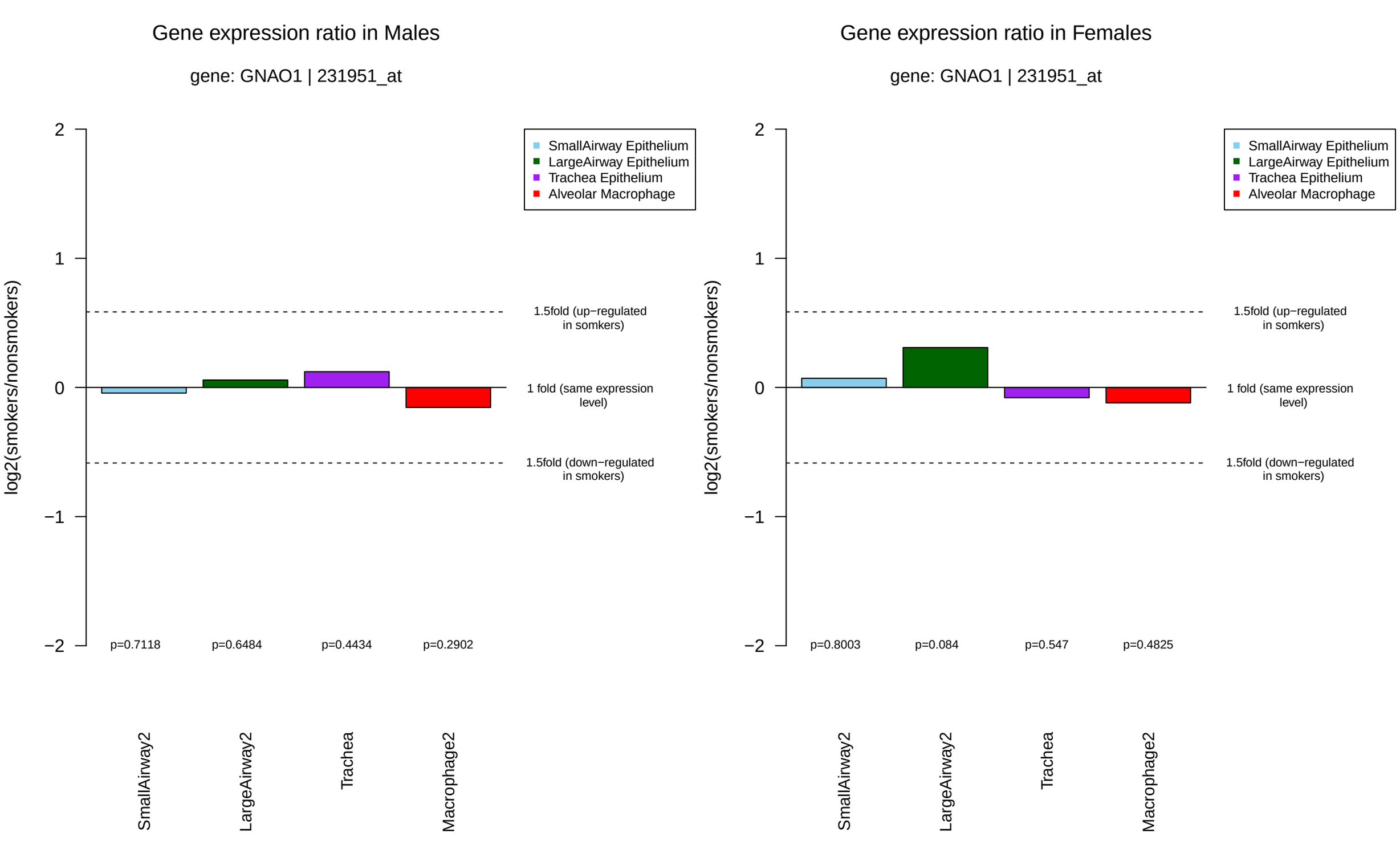
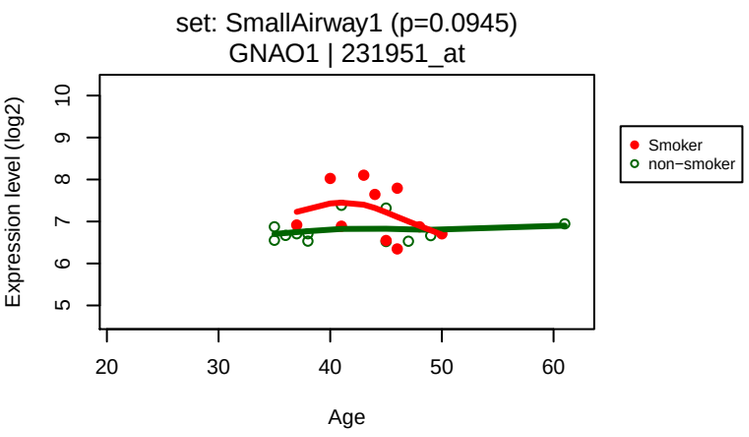
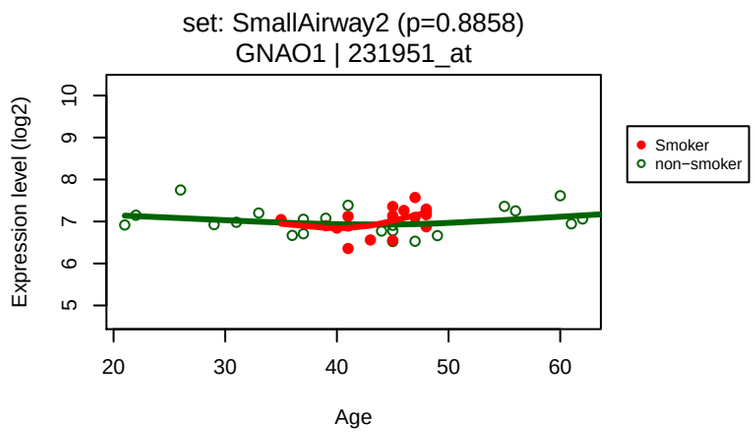
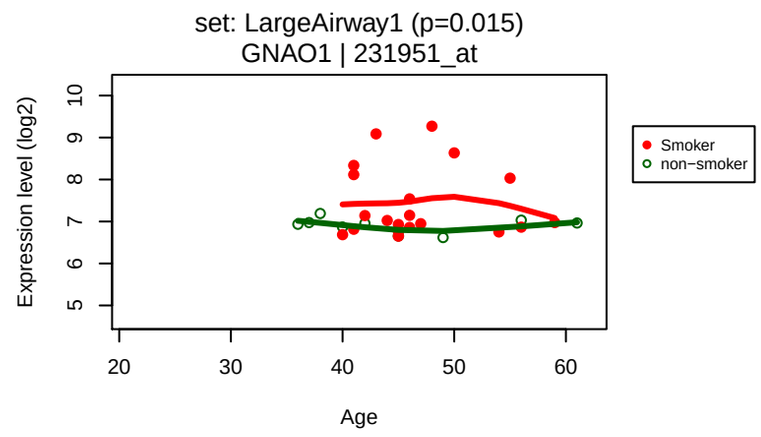

Gene expression of smokers and nonsmokers
gene: GNAO1 | 231951_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0945
p=0.8858
p=0.2694
p=0.015
p=0.1211
p=0.6964
p=0.8146
p=0.1901
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

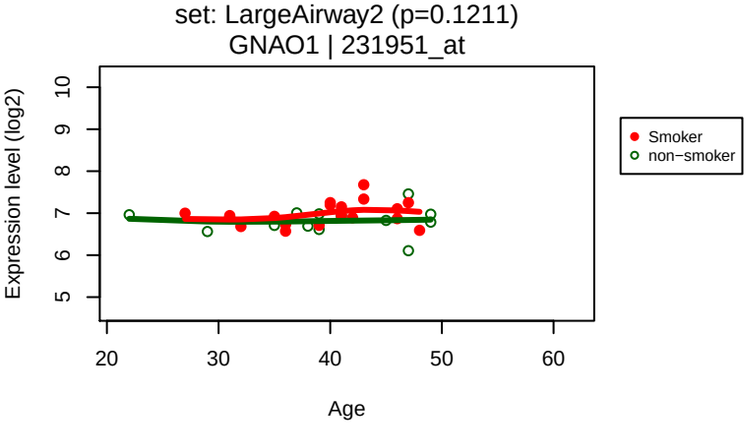

Gene expression ratio (smokers/nonsmokers)
gene: GNAO1 | 231951_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0945
p=0.8858
p=0.2694
p=0.015
p=0.1211
p=0.6964
p=0.8146
p=0.1901
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

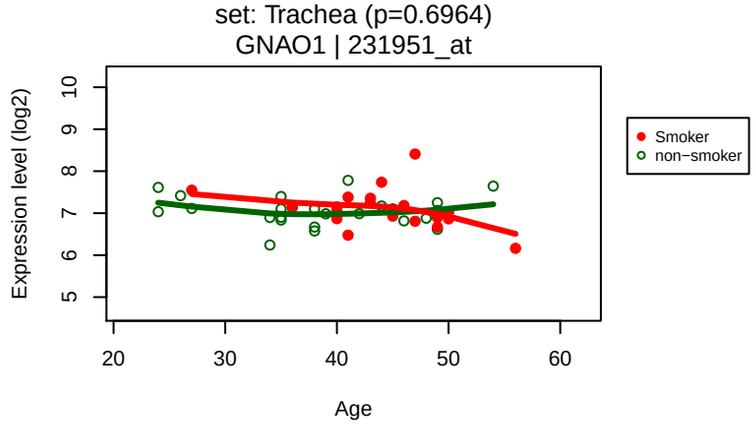

Gene expression ratio in Males
gene: GNAO1 | 231951_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7118
p=0.6484
p=0.4434
p=0.2902
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAO1 | 231951_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8003
p=0.084
p=0.547
p=0.4825
SmallAirway2
LargeAirway2
Trachea
Macrophage2

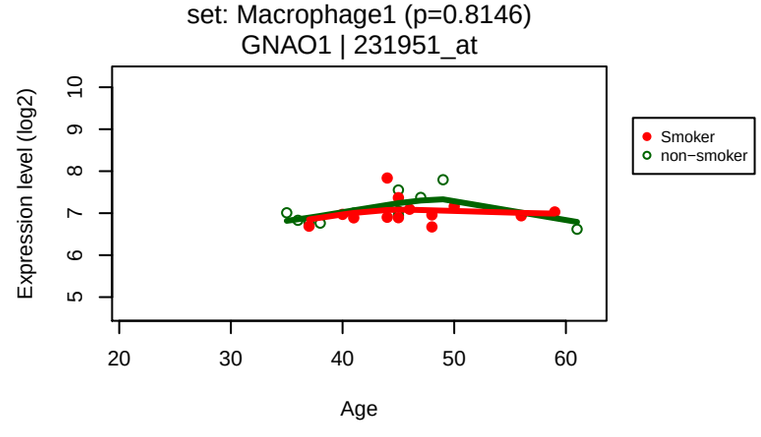

set: SmallAirway1 (p=0.0945)
GNAO1 | 231951_at
Expression level (log2)
Age
Smoker
non−smoker

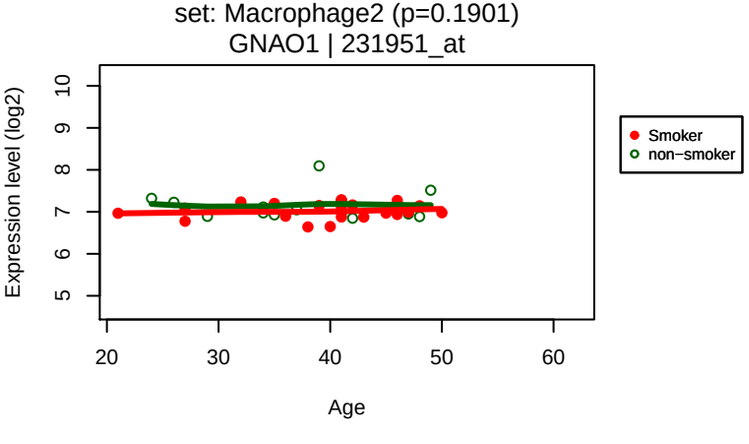

set: SmallAirway2 (p=0.8858)
GNAO1 | 231951_at
Expression level (log2)
Age
Smoker
non−smoker

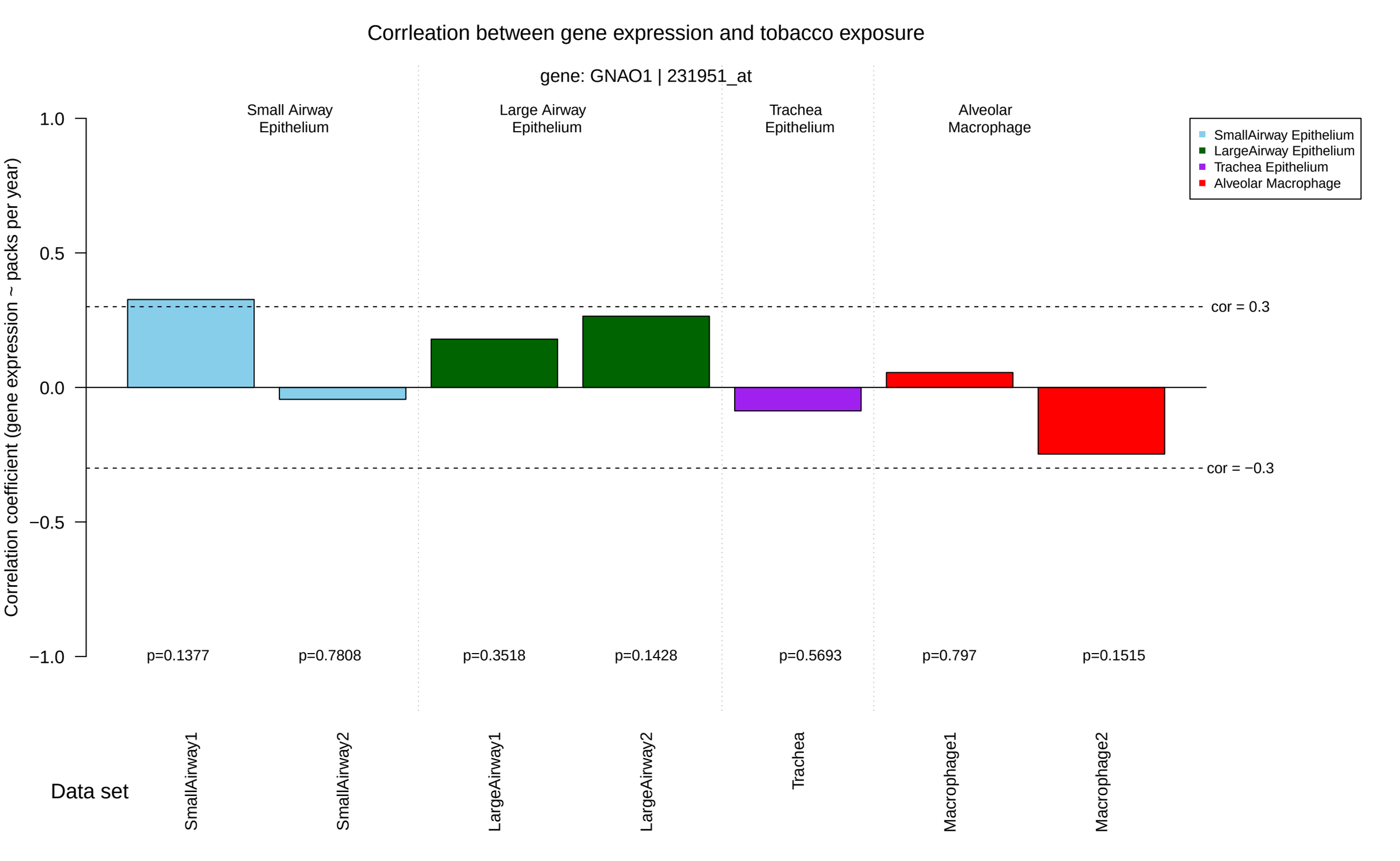

set: LargeAirway1 (p=0.015)
GNAO1 | 231951_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1211)
GNAO1 | 231951_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6964)
GNAO1 | 231951_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8146)
GNAO1 | 231951_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1901)
GNAO1 | 231951_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAO1 | 231951_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1377
p=0.7808
p=0.3518
p=0.1428
p=0.5693
p=0.797
p=0.1515
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2